

Sequence Detector

Overlapping

non
overlapping

⑩ overlapping sequence detector:

detected 1001

i/p: 1 0 0 1 0 0 1 0 0 1 0 0 1 1 1 0 1 1

o/p.

Sequence Detector

Overlapping

non
overlapping

⑩ overlapping sequence detector:

detected 1001

100100100100111011

o/p.

↓
0

Sequence Detector

Overlapping

non overlapping

⑩ overlapping sequence detector:

detect 1001

1 0 0 1 0 0 1 0 0 1 0 0 1 1 1 0 1 1

o/p. 0 0

Sequence Detector

Overlapping

non overlapping

⑩ overlapping sequence detector:

detected 1001

1 0 0 1 0 0 1 0 0 1 0 0 1 1 1 0 1 1

o/p.
0 0 0

Sequence Detector

Overlapping

non
overlapping

⑩ overlapping sequence detector:

detect 1001

i/p: 1 0 0 1 0 0 1 0 0 1 0 0 1 1 1 0 1 1
 ↓
o/p: 0 0 0 1

Sequence Detector

Overlapping

non overlapping

⑩ overlapping sequence detector:

detect 1001

1 0 0 1 0 0 1 0 0 1 0 0 1 1 1 0 1 1

o/p.

0 0 0 1 0

Sequence Detector

Overlapping

non overlapping

⑩ overlapping sequence detector:

detect 1001

1 0 0 1 0 0 1 0 0 1 0 0 1 1 1 0 1 1

o/p.

0 0 0 1 0 0

Sequence Detector

Overlapping

non overlapping

⑩ overlapping sequence detector:

detected 1001

1 0 0 1 0 0 1 0 0 1 0 0 1 1 1 0 1 1

o/p.

0 0 0 1 0 0 1

Sequence Detector

Overlapping

non
overlapping

⑩ overlapping sequence detector:

detected 1001

1 0 0 1 0 0 1 0 0 1 0 0 1 1 1 0 1 1

o/p.

0 0 0 1 0 0 1 0 0 1 0 0 1 0 0 0 0 0

Sequence Detector

Overlapping

non overlapping

① overlapping sequence detector:

detect 1001

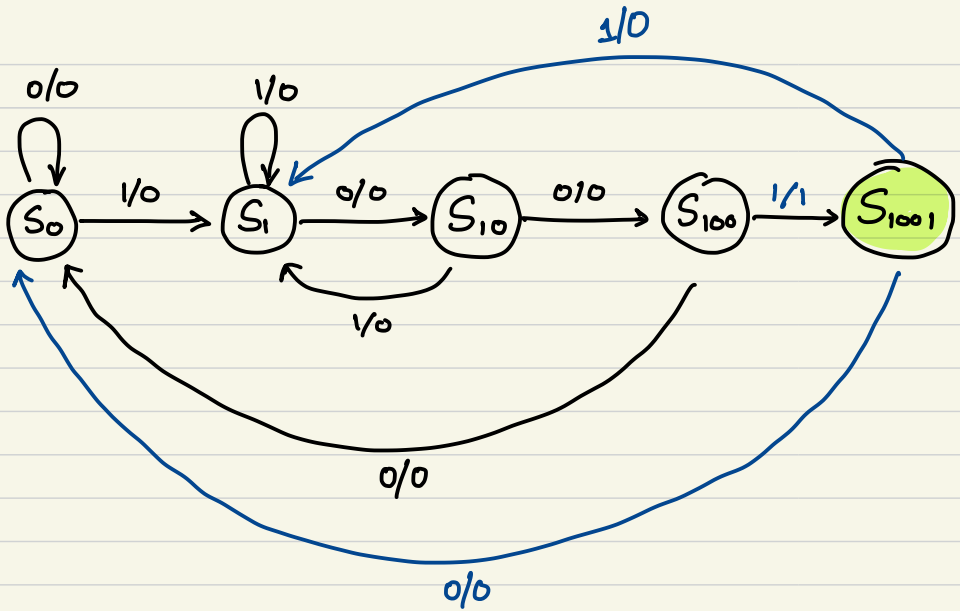
1 0 0 1 0 0 1 0 0 1 0 0 1 1 1 0 1 1

o/p. 0 0 0 1 0 0 1 0 0 1 0 0 1 0 0 0 0 0

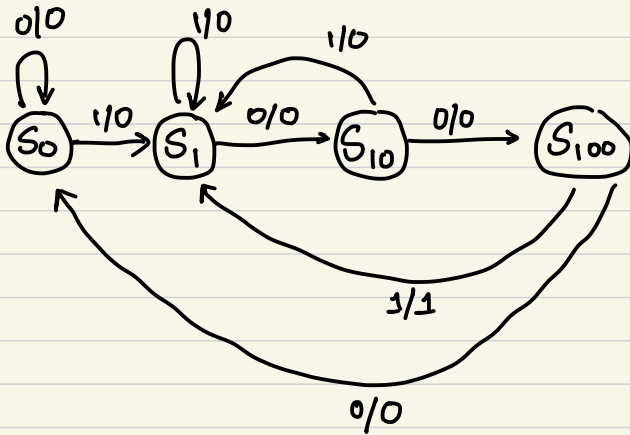
Non-overlapping Sequence Detector:

Input: 1 0 0 1 0 0 1 0 0 1 0 0 1 1 1 0 1 1
 ↓ ↓
Output 0 0 0 1 0 0 0 0 0 1 0 0 0 0 0 0 0 0

non-overlapping



overlapping



State table (overlapping)

$S_0 \rightarrow 00$

$S_1 \rightarrow 01$

$S_{10} \rightarrow 10$

$S_{100} \rightarrow 11$

present	next		output	
	$x=0$	$x=1$	$x=0$	$x=1$
S_0	S_0	S_1	0	0
S_1	S_{10}	S_1	0	0
S_{10}	S_{100}	S_1	0	0
S_{100}	S_0	S_1	0	1

State table (Overlapping)

present	next		output	
	$x=0$	$x=1$	$x=0$	$x=1$
S_0	S_0	S_1	0	0
S_1	S_{10}	S_1	0	0
S_{10}	S_{100}	S_1	0	0
S_{100}	S_0	S_1	0	1

$$S_0 \rightarrow 00$$

$$S_1 \rightarrow 01$$

$$S_{10} \rightarrow 10$$

$$S_{100} \rightarrow 11$$

Merged State Table

x	Q_1	Q_0	Q_1^+	Q_0^+	Y
0	0	0	0	0	0
	0	1	1	0	0
	1	0	1	1	0
	1	1	0	0	0
1	0	0	0	1	0
	0	1	0	1	0
	1	0	0	1	0
	1	1	0	1	1

Merged State Table

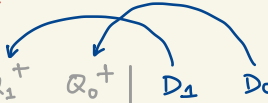
x	Q_1	Q_0	Q_1^+	Q_0^+	Y
0	0	0	0	0	0
	0	1	1	0	0
	1	0	1	1	0
	1	1	0	0	0
1	0	0	0	1	0
	0	1	0	1	0
	1	0	0	1	0
	1	1	0	1	1

Excitation Table:



x	Q_1	Q_0	Q_1^+	Q_0^+	D_1	D_0	Y
0	0	0	0	0	0	0	0
	0	1	1	0	1	0	0
	1	0	1	1	1	1	0
	1	1	0	0	0	0	0
1	0	0	0	1	0	1	0
	0	1	0	1	0	1	0
	1	0	0	1	0	1	0
	1	1	0	1	0	1	1

Excitation Table:

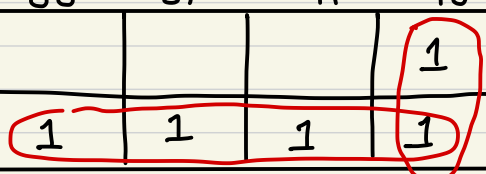


x	Q_1	Q_0	Q_1^+	Q_0^+	D_1	D_0	Y
0	0	0	(S ₀)	0	0	0	0
	0	1	(S ₁)	0	1	0	0
	1	0	(S ₁₀)	1	1	1	0
	1	1	(S ₁₀₀)	0	0	0	0
1	0	0		0	1	0	0
	0	1		0	1	1	0
	1	0		0	1	1	0
	1	1		0	1	1	1

$$D_1 = \bar{x} \bar{Q}_1 Q_0 + \bar{x} Q_1 \bar{Q}_0$$

$$= \boxed{\bar{x} (Q_1 \oplus Q_0)}$$

D₀



$x \backslash Q_1 Q_0$	00	01	11	10
0				1
1	1	1	1	1

$$D_0 = x + Q_1 \bar{Q}_0$$

$$Y = x Q_1 Q_0$$

Step 5
Circuit Diagram

